

Siddhartha Prasad

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I am a programming languages researcher interested in the human aspects of programming and formal reasoning. I study not only how PL and FM techniques can solve problems programmers face, but also how programmers' experiences can inform the design of our languages, logics, and software tools. My research is informed by my time as an engineer: I have written code that doesn't do what I want it to, and I want to spare everyone else the indignity.

EDUCATION

Brown University

Providence, RI Jan 2022 – present

Ph.D. in Computer Science

Tufts University

Medford, MA May 2016

B.S. in Computer Science and Mathematics

Summa cum Laude; High Thesis Honors

EMPLOYMENT

Apple

Seattle, WA

Research Intern · Human-Centered Machine Learning

May – Sept 2023

Explored how AI and formal-methods techniques can support early childhood education.

Microsoft

Redmond, WA

Software Engineer II · Applied AI

April 2018 – Sept 2021

- Developer on the Azure Cognitive Services team, which lets developers embed AI into apps, bots, and websites.
- Built a containerized framework that allowed AI models to be run portably across operating systems and architectures.

Software Engineer · Developer Ecosystem Platform

Aug 2016 – March 2018

- Developed APIs and features for XAML, a UI language used to build Universal Windows apps.
- Designed XAML APIs for input modalities (keyboard, gamepad, ink), accessibility, and cross-platform language support.

Software Engineering Intern · Developer Ecosystem

May – Aug 2015

INRIA

Saclay, France

Research Intern · Parsifal

June – July 2014

Research on correctness certificates for first-order term rewriting.

PUBLICATIONS

Diagramming Program Values by Spatial Refinement, **PLDI 2026**

Siddhartha Prasad, Michael Tu, Karan Kashyap, Tim Nelson, Shriram Krishnamurthi

Meaningful Human-in-the-Loop Checking of GenAI Synthesis for Restricted Languages, **ECOOP 2026**

Siddhartha Prasad, Skyler Austen, Kathi Fisler, Shriram Krishnamurthi

hS: Speculative Script Reordering at Subprocess Granularity, **OSDI 2026**

Georgios Liargkovas, Di Jin, Tianyu (Ezri) Zhu, Dan Liu, A. Bolun Thompson, Anirudh Narsipur, Seong-Heon Jung, *Siddhartha Prasad*, Diomidis Spinellis, Michael Greenberg, Konstantinos Kallas, Nikos Vasilakis

A Misconception-Driven Adaptive Tutor for Linear Temporal Logic, **CAV 2025**

[Distinguished Paper]

Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi

Lightweight Diagramming for Lightweight Formal Methods: A Grounded Language Design, **ECOOP 2025**

[Distinguished Paper; Distinguished Artifact]

Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi

Misconceptions in Finite-Trace and Infinite-Trace Linear Temporal Logic, **FM 2024**

Ben Greenman, *Siddhartha Prasad*, Antonio Di Stasio, Shufang Zhu, Giuseppe De Giacomo, Shriram Krishnamurthi, Marco Montali, Tim Nelson, Milda Zizyte

ContextQ: Generated Questions to Support Meaningful Parent-Child Dialogue While Co-Reading, **IDC 2024**

Griffin Dietz Smith, *Siddhartha Prasad*, Matt J Davidson, Leah Findlater, R Benjamin Shapiro

Forge: A Tool and Language for Teaching Formal Methods, **OOPSLA 2024**

Tim Nelson, Ben Greenman, *Siddhartha Prasad*, Tristan Dyer, Ethan Bove, Qianfan Chen, Charles Cutting, Thomas Del Vecchio, Sidney LeVine, Julianne Rudner, Ben Ryjikov, Alexander Varga, Andrew Wagner, Luke West, Shriram Krishnamurthi

Conceptual Mutation Testing for Student Programming Misconceptions, **«Programming» 2024**

Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi

Generating Programs Trivially: Student Use of Large Language Models, **CompEd 2023**

Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi

Making Hay from Wheats: A Classsourcing Method to Identify Misconceptions, **Koli Calling 2022**

Siddhartha Prasad, Ben Greenman, Tim Nelson, John Wrenn, Shriram Krishnamurthi

Large-Scale Intelligent Microservices, **IEEE BigData 2020**

Mark Hamilton, Nick Gonsalves, Christina Lee, Anand Raman, Brendan Walsh, *Siddhartha Prasad*, Dalitso Banda, Lucy Zhang, Lei Zhang, William T Freeman

TEACHING

Teaching Assistant, **Brown University**

Logic for Systems

2022–2026

Programming Languages

2025

Teaching Assistant, **Tufts University**

Programming Languages

2015–2016

Algorithms

2014

SERVICE

2026 · Artifact Reviewer, The Art, Science, and Engineering of Programming

AFFILIATIONS & CERTIFICATIONS

Sheridan Teaching Seminar Certificate · Brown University
Tau Beta Pi, The Engineering Honor Society · Tufts University